

DANA 4820 Project Part 1

2026-07-01

Course: DANA 4820

Team members: Nandini Bhatia, Diego Alberto, Khushpreet Kaur, Haruka Sawakami

Part (a)

Summary of the Beyond Multiple Linear Regression, Chapter 4

4. Poisson Regression

4.1 Learning Objectives

The primary objective of this section is to understand the foundational concepts of Poisson regression and recognize why Simple Linear Regression models are inadequate for modeling count data. Through this chapter, we aim to grasp the model's assumptions for inference, the estimation and interpretation of coefficients using R, and the methods for comparing models through hypothesis testing.

4.2 Introduction to Poisson Regression

Poisson regression is specifically designed to model response variables (Y) that represent counts, that is the average number of occurrences per unit of time or space. Examples out of four include the annual number of motorcycle deaths in a state or the number of deformed fish in Minnesota lakes over the last decade. Count data has unique characteristics: they are discrete, can take only positive number and theoretically unbounded above (from 0 to ∞).

While it might seem to be reasonable to apply a standard Linear Least Squares Regression (LLSR) model, such as $\lambda_i = \beta_0 + \beta_1 x_i$, doing so introduces two critical violations of regression assumptions:

1. Violation of Boundedness (Negative Values):

A linear regression line extends infinitely and is guaranteed to predict negative values for certain values of x_i . However, the Poisson mean rate parameter (λ_i) represents an average count and can only take non-negative values.

2. Violation of Homoscedasticity (Equal Variance):

In LLSR, the variance of residuals is assumed to be constant across all levels of X . For Poisson data, however, as the mean rate (λ) increases, the variability of the data also increases. This is because a key property of the Poisson distribution is that its mean is equal to its variance ($E(Y) = Var(Y) = \lambda$).

To resolve these issues, instead of modeling λ_i directly, we express the natural logarithm of the mean rate, $\log(\lambda_i)$, as a linear function:

$$\log(\lambda_i) = \beta_0 + \beta_1 x_i$$

Taking the exponential of both sides keeps the predicted mean rate λ_i positive for any value of x_i , which satisfies the mathematical constraint that count data cannot be negative.

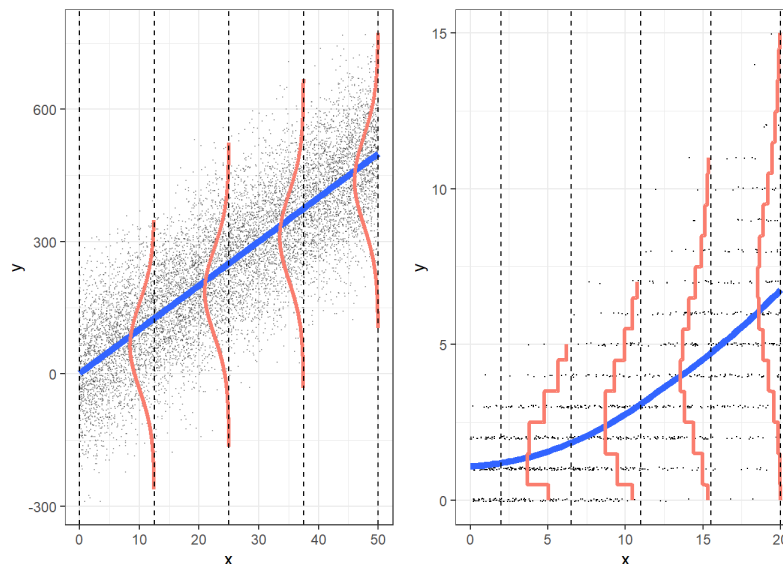
Furthermore, unlike the LLSR, the Poisson regression model does not contain a separate error term (ϵ). In the LLSR, ϵ represents the random variation around the mean. In Poisson regression, a separate error term is unnecessary because the single parameter λ_i completely determines both the mean and the variance of the response variable ($Y_i \sim \text{Poisson}(\lambda_i)$).

4.2.1 Poisson Regression Assumptions

For a Poisson regression model to use, the following four key assumptions must be satisfied:

- Poisson Response:** The response variable must be a count of occurrences that follows a Poisson distribution within a defined unit of time or space.
- Independence:** The observations must be independent of one another.
- Mean = Variance:** A fundamental property of the Poisson distribution is that its mean is equal to its variance ($E(Y) = Var(Y) = \lambda$). We must check if the means and variances are approximately equal across groups of the explanatory variable.
- Linearity:** The log of the mean rate, $\log(\lambda)$, must be a linear function of the explanatory variables (X).

4.2.2 A Graphical Look at Poisson Regression



Regression models: Linear regression (left) and Poisson regression (right).

Figure illustrates a comparison of the LLSR model for inference to Poisson regression using a log function of λ .

Before fitting a Poisson regression model, we should visually evaluate our data to see if the assumptions hold. In standard linear regression, we look for a straight line in a scatterplot. However, for a Poisson regression model:

1. **We do not expect a linear relationship between X and the raw counts Y .** Instead, if the relationship follows a Poisson distribution, a scatterplot of the explanatory variable (X) against the response variable (Y) should show an exponential curve rather than a straight line.
2. **We look for a linear relationship between X and the log of the mean response.** To visually check the linearity assumption ($\log(\lambda) = \beta_0 + \beta_1 X$), we can break a continuous explanatory variable into discrete groups and calculate the mean for each group.
3. **The plot of the log means against X should look approximately linear.** If the linearity assumption is satisfied, plotting the natural logarithm of these means against the average X values for each group should show a reasonably straight line.

4.3 Case Study Overview

In this section, we overview the three case studies introduced in the textbook, each based on real data and real questions to illustrate different aspects of Poisson regression modeling:

1. **Household Size in the Philippines:** Introduces the basic idea of regression with a Poisson response along with its structural assumptions, adding a quadratic term to determine an optimal size, and methods of model comparison.
2. **Campus Crime:** Introduces offsets to account for sampling effort and overdispersion when actual variability exceeds what is expected.
3. **Weekend Drinking:** Uses a modification of a Poisson model to account for more zeros than expected.

Note: Since the topics covered with campus crime and weekend drinking case studies are beyond the scope of our course, we will focus exclusively on the first case study: Household Size in the Philippines.

4.4 Case Study: Household Size in the Philippines

This case study explores how household size varies across different demographics in the Philippines, illustrating how international agencies utilize such data to determine population needs.

The analysis is guided by two primary questions: 1. *At what age are heads of households in the Philippines most likely to find the largest number of people in their household?* 2. *Is this association between household size and the head's age similar for poorer households?* To answer these questions, we examine the relationship between household size and the age of the household head, while controlling for location and income level.

4.4.1 Data Organization

The data used in this study is a subset of 1,500 observations sampled from the 40,000 total observations in the 2015 Family Income and Expenditure Survey (FIES), which is conducted every three years by the Philippine Statistics Authority (PSA). In this data set, the variables and defined as follows:

Response Variable (Y)

- **total** : The number of household **members other than the head of the household**. Because this represents a discrete count, it is modeled as a Poisson response where the parameter of interest is λ (the average number of non-head members per household).

Explanatory Variable (X)

- **age** : The age of the head of the household (continuous variable).

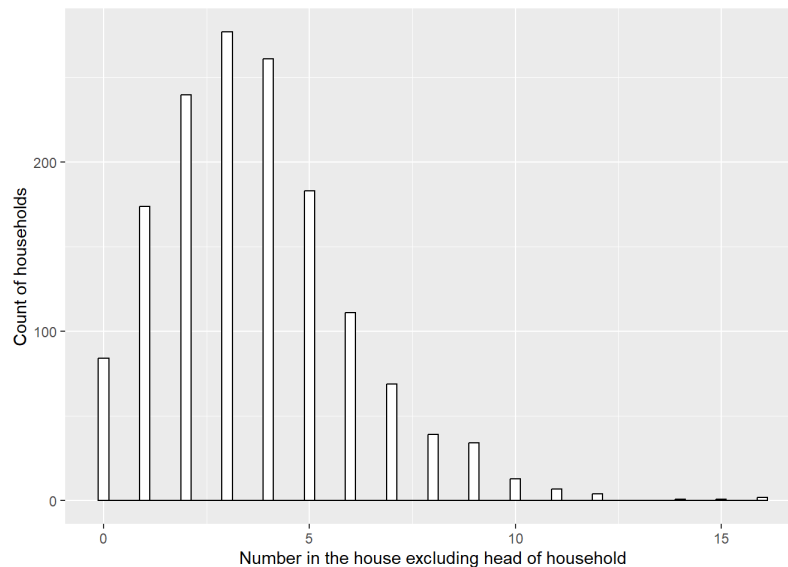
- **roof** : The type of roof material, categorized into predominantly light/salvaged material (used as a proxy to measure poorer households) or predominantly strong material (categorical variable).
- **location** : The geographic five regions (Central Luzon, Davao Region, Ilocos Region, Metro Manila, and Visayas) of the household (categorical variable).
- **numLT5** : The number of household members under 5 years old (discrete count variable).

The first five observations from the Philippines Household case study, our data set `fHH1.csv` .

...	1	location	age	total	numLT5	roof
1	CentralLuzon	65	0	0	Predominantly Strong Material	
2	MetroManila	75	3	0	Predominantly Strong Material	
3	DavaoRegion	54	4	0	Predominantly Strong Material	
4	Visayas	49	3	0	Predominantly Strong Material	
5	MetroManila	74	3	0	Predominantly Strong Material	

4.4.2 Exploratory Data Analyses

In this section, we perform Exploratory Data Analysis (EDA) to examine the descriptive statistics of the dataset and visually evaluate the data to verify the four assumptions of Poisson regression. The household size (excluding the head) ranges from 0 to 16, with an overall mean of 3.68 and a variance of 5.53. By Roof Material, over 11.1% of households are made from predominantly light/salvaged material. The mean household size is highly similar between groups: 3.69 (variance of 5.55) for strong roofs, and 3.64 (variance of 5.41) for light/salvaged roofs. By Location, the Visayas region has the largest average household size mean of 3.90, while the Davao Region has the smallest mean of 3.39.

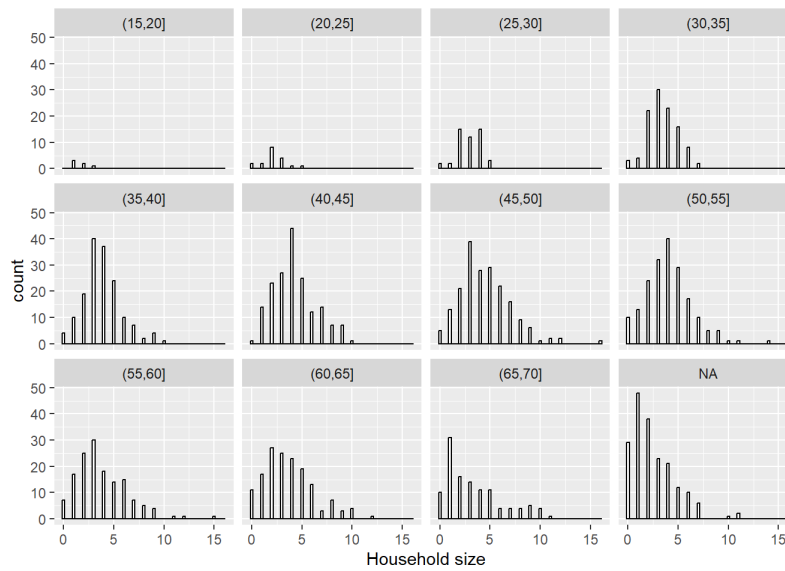


Distribution of household size in 5 Philippine regions.

Step by Step Checking Assumptions:

- **Assumption 1: Poisson Response**

The overall histogram of household size is clearly right-skewed, confirming the data is not normally distributed. When grouped by the primary explanatory variable `age` of the household head, the distributions behave reasonably like discrete counts, suggesting that the Poisson Response assumption is satisfactory.



Distribution of household sizes by age group of the household head.

- **Assumption 2: Independence**

Due to a lack of detailed background information regarding the sampling process and household relationships in this subset, a formal verification of the Independence assumption is not possible. If households were clustered (e.g., selected by village with shared customs) rather than sampled individually at random, independence would be violated. However, we proceed assuming independence holds.

- **Assumption 3: Mean = Variance**

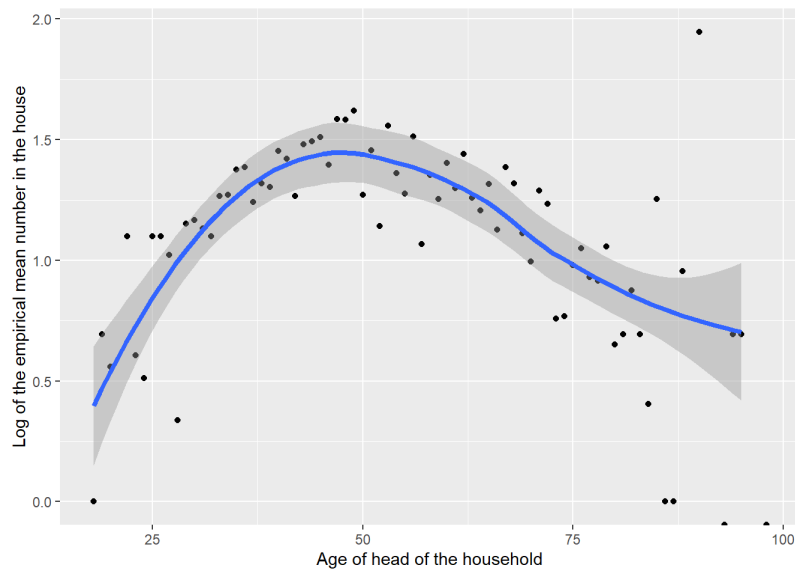
To evaluate whether the mean equals the variance, the empirical means and variances of household sizes are calculated across 5-year age intervals. While variance generally increases with age, it is smaller than the mean at younger ages and larger at older ages. This potentially evidents a violation of mean = variance, though the authors note that these violations are modest for now.

Compare mean and variance of household size within each age group.

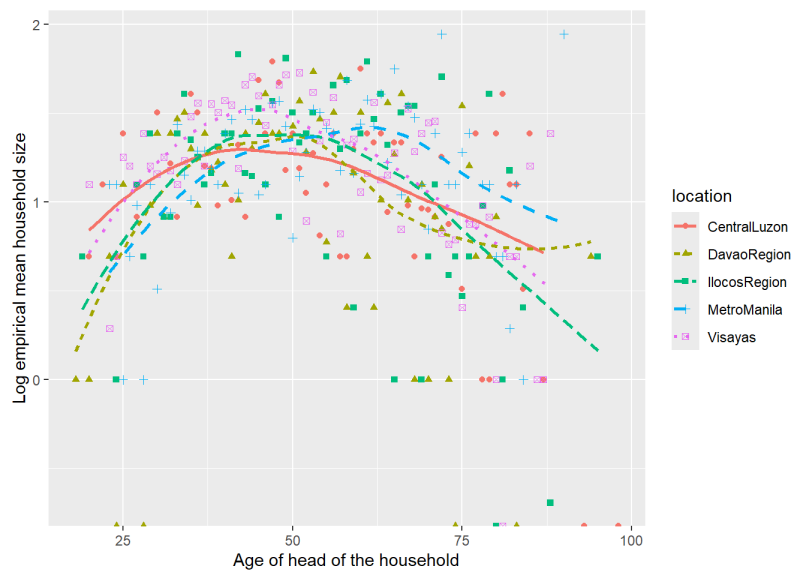
Age Groups	Mean	Variance	n
(15,20]	1.666667	0.6666667	6
(20,25]	2.166667	1.5588235	18
(25,30]	2.918367	1.4098639	49
(30,35]	3.444444	2.1931464	108
(35,40]	3.841772	3.5735306	158
(40,45]	4.234286	4.4447947	175
(45,50]	4.489691	6.3962662	194
(50,55]	4.010638	5.2512231	188
(55,60]	3.806897	6.5318966	145
(60,65]	3.705882	6.1958204	153
(65,70]	3.339130	7.9980168	115
NA	2.549738	5.5435657	191

- **Assumption 4: Linearity**

Because the true mean rate λ_i is unknown, we cannot directly assess linearity on the log scale using standard scatterplots. To resolve this, we utilize **empirical log means** against age. The resulting plot reveals a curved trend rather than a straight line, indicating that a linear term alone is insufficient and a **quadratic term** age^2 should be considered. In addition, plotting separate curves for each region shows a highly similar trend across locations. If these trends differed significantly, an age-by-region interaction term would be required.



The log of the mean household sizes, besides the head of household, by age of the head of household, with loess smoother.



Empirical log of the mean household sizes vs. age of the head of household, with loess smoother by region.

4.4.3 Estimation and Inference

Based on the previous EDA, we begin our formal model evaluation by fitting a baseline linear Poisson model, where $\log(\lambda)$ is modeled as a linear function of age . The estimated regression equation reported by R is:

$$\log(\hat{\lambda}) = 1.55 - 0.0047age$$

```
modela = glm(total ~ age, family = poisson, data = fHH1)
```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) 1.549942225 0.0502754106 30.829032 1.070156e-208
## age        -0.004705881 0.0009363388 -5.025832 5.012548e-07
```

```
## Residual deviance = 2337.089 on 1498 df
## Dispersion parameter = 1
```

Interpretation of Coefficients:

In Poisson regression, because the response variable is logged, we interpret the effects multiplicatively rather than additively. By comparing the model at age x and age $x + 1$, we find that exponentiating the coefficient (e^{β_1}) yields the **multiplicative factor** (also known as the **rate ratio** or **relative risk**).

$$\begin{aligned}
\log(\lambda_X) &= \beta_0 + \beta_1 X \\
\log(\lambda_{X+1}) &= \beta_0 + \beta_1 (X + 1) \\
\log(\lambda_{X+1}) - \log(\lambda_X) &= \beta_1 \\
\log\left(\frac{\lambda_{X+1}}{\lambda_X}\right) &= \beta_1 \quad (\#eq : rateRatio) \\
\frac{\lambda_{X+1}}{\lambda_X} &= e^{\beta_1}
\end{aligned}$$

Here, $e^{-0.0047} = 0.995$. This means that with each additional year of age of the household head, the mean household size decreases by 0.5% (since $1 - 0.995 = 0.005$). Conversely, a 1-year decrease in age predicts a 0.47% increase in size. Generally, logged variables express multiplicative effects on the original scale, whereas unlogged variables express usual additive effects.

Confidence Intervals (CI) using Standard Error (SE):

R output provides a Standard Error (SE=0.00094) for the age coefficient, allowing us to construct a 95% Confidence Interval. It is best practice to construct the confidence interval for the raw coefficient first and then exponentiate the endpoints. This is because the estimated coefficients ($\hat{\beta}_1$) follow a normal distribution much more closely than the exponentiated coefficients. Resulting Interval is that exponentiating the endpoints yields a multiplicative factor interval of (e-0.0065, e-0.0029)=(0.993, 0.997). We are 95% confident that the mean household size decreases between 0.7% and 0.3% for each additional year.

$$\begin{aligned}
&(\hat{\beta}_1 - Z^* \cdot SE(\hat{\beta}_1), \quad \hat{\beta}_1 + Z^* \cdot SE(\hat{\beta}_1)) \\
&(-0.0047 - 1.96 * 0.00094, \quad -0.0047 + 1.96 * 0.00094) \\
&(-0.0065, -0.0029).
\end{aligned}$$

Hypothesis Testing and Significance:

To test whether `age` is significantly associated with household size ($H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$), we evaluate it in two ways:

```
# CI for betas using profile Likelihood
confint(modela)
```

```
##                2.5 %      97.5 %
## (Intercept)  1.451170100  1.648249185
## age         -0.006543163 -0.002872717
```

```
exp(confint(modela))
```

```
##                2.5 %      97.5 %
## (Intercept)  4.2681057  5.1978713
## age         0.9934782  0.9971314
```

1. Confidence Interval Approach: The 95% CI for the multiplicative factor (e^{β_1}) is (0.993, 0.997), which does not include 1 (likewise, the 95% CI for the raw coefficient β_1 does not include 0). Because 1 represents “no change” on the multiplicative scale, excluding 1 indicates that the model with `age` is significantly preferred over an intercept-only model.

2. Wald-type Test: A Wald-type test statistic is calculated by dividing the estimated coefficient by its SE. Under a sufficiently large sample size (n), this test statistic follows a standard normal distribution when H_0 is true:**

$$Z = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{-0.0047}{0.00094} = -5.026$$

The two-sided p-value is nearly zero ($p = 0.000000501$). This provides statistically overwhelming evidence ($Z = -5.026, p < .001$) that the average household size significantly decreases as the age of the household head increases.

4.4.4 Using Deviances to Compare Models

Our objective here is to assess if the age is a significant predictor for the household size, a drop in deviance test was conducted to compare the null model to the model incorporating age as linear predictor.

The hypotheses are formally stated as follows:

$$\begin{aligned}
H_0 : \log(\lambda) &= \beta_0 \quad (\beta_1 = 0, \text{ age is not a significant predictor}) \\
H_A : \log(\lambda) &= \beta_0 + \beta_1 \text{age} \quad (\beta_1 \neq 0, \text{ age is a significant predictor})
\end{aligned}$$

```
# Null model
model0 <- glm(total ~ 1, family = poisson, data = fHH1)

# Model with age (Linear)
modela <- glm(total ~ age, family = poisson, data = fHH1)

# Drop-in-deviance test
drop_in_dev_1 <- anova(model0, modela, test = "Chisq")
drop_in_dev_1
```

```
## Analysis of Deviance Table
##
## Model 1: total ~ 1
## Model 2: total ~ age
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      1499      2362.5
## 2      1498      2337.1  1    25.399 4.661e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The null model, which contains no predictors, has a residual deviance of 2362 on 1499 degrees of freedom. When included the age as a predictor, the residual deviance decreased to 2337 on 1498 degrees of freedom, this results in a drop in deviance of 25.4 with one degree of freedom. Then we evaluate this against a chi-square distribution with one degree of freedom, resulting in a very tiny p-value (4.66×10^{-7}) that is below the significance threshold of 0.05 Interpretation: We reject H_0 . There is statistically significant evidence that age is a meaningful predictor of household size, and therefore the model with age is preferred over the null model

4.4.6 Second Order Model

We already showed that age matters for household size. Then when we looked at our exploratory graphs earlier (section 4.4.2), the relationship wasn't a straight line - it went up, peaked and then came down. To determine whether the quadratic term significantly improves model fit, a drop-in-deviance test is conducted comparing the linear model to the quadratic model. The hypotheses are formally stated as follows:

$$H_0 : \log(\lambda) = \beta_0 + \beta_1 \text{age} \quad (\beta_2 = 0, \text{quadratic term is not needed})$$

$$H_A : \log(\lambda) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{age}^2 \quad (\beta_2 \neq 0, \text{quadratic term is needed})$$

```
# Add quadratic term
fHH1 <- fHH1 %>% mutate(age2 = age * age)

# Fit quadratic model
modela2 <- glm(total ~ age + age2, family = poisson, data = fHH1)
summary(modela2)
```

```
##
## Call:
## glm(formula = total ~ age + age2, family = poisson, data = fHH1)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.325e-01  1.788e-01  -1.859   0.063 .
## age          7.089e-02  6.890e-03  10.288 <2e-16 ***
## age2        -7.083e-04  6.406e-05 -11.058 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 2362.5  on 1499  degrees of freedom
## Residual deviance: 2200.9  on 1497  degrees of freedom
## AIC: 6579.8
##
## Number of Fisher Scoring iterations: 5
```

```
# Drop-in-deviance test: Linear vs quadratic
drop_in_dev_2 <- anova(modela, modela2, test = "Chisq")
drop_in_dev_2
```

```
## Analysis of Deviance Table
##
## Model 1: total ~ age
## Model 2: total ~ age + age2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1498      2337.1
## 2      1497      2200.9  1   136.15 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Calculate optimal age
beta1 <- coef(modela2)["age"]
beta2 <- coef(modela2)["age2"]
optimal_age <- -beta1 / (2 * beta2)
cat("Optimal age:", round(optimal_age, 2))
```

```
## Optimal age: 50.04
```

```
# Maximum estimated household size
max_log_total <- coef(modela2)[1] + beta1 * optimal_age +
  beta2 * optimal_age^2
cat("Max mean household size:", round(exp(max_log_total), 3))
```

```
## Max mean household size: 4.225
```

$$\log(\text{mean numHouse}) = -0.333 + 0.071 \cdot \text{age} - 0.00071 \cdot \text{age}^2$$

The positive coefficient associated with the linear age ($\hat{\beta}_1 = 0.071$, $p < 0.001$) indicates that household size initially increases with the age of the household head. The negative coefficient associated with the quadratic term ($\hat{\beta}_2 = -0.00071$, $p < 0.001$) confirms the presence of a concave relationship, in other words household size reaches a maximum at a certain age before subsequently declining. We can expect that household size grows as children are born and raised, then decreases as children leave the household.

We then asked: does adding age^2 actually make the model better or not? We compared the simpler model to the more complex $\text{model}(\text{age} + \text{age}^2)$. The deviance dropped by 136.1 and the p-value is almost zero. Therefore we reject H_0 and conclude that the inclusion of the quadratic term significantly improves model fit, and the quadratic model is strongly preferred over the linear model.

To determine the age at which household size is estimated to be at its maximum, the derivative of the fitted model with respect to age was set equal to zero:

$$\frac{d}{d(\text{age})} \log(\text{total}) = 0.071 - 2(0.00071) \cdot \text{age} = 0$$

$$\text{age} = 0.071 / 0.00142 \approx 50.04 \text{ years}$$

To find exactly where the peak is, we use a basic calculus trick: at the very top of a curve, the slope is flat (zero). So we take the derivative of our equation, set it to zero, and solve for age. That gives us $\text{age} \approx 50.04$.

Then we plug that age back into the original equation to find out how big the household is at that peak. That gives us a log value of 1.441, and converting that back to a normal number (using $e^{1.441}$) gives us about 4.23 people.

Interpretation: Household size peaks when the head of household is around 50 years old, with about 4.2 extra people living in the home at that point — then it starts declining again.

4.4.7 Adding a Covariate

So far our model only uses age to predict household size. But it's reasonable to think that where someone lives might also affect how many people are in their household. So we wanted to test: after we've already accounted for age, does location also matter?

Location isn't a number, it's a category (Central Luzon, Davao, Ilocos, Metro Manila, Visayas), so R can't just drop it into the equation as-is. Instead, R picks one category as the "baseline" to compare everything else to — here it picked Central Luzon — and creates a 0/1 indicator for each of the other four regions. So basically, the model is asking "compared to Central Luzon, is this region higher or lower?" for each region separately. The hypotheses are formally stated as follows:

$$H_0 : \log(\lambda) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{age}^2 \quad (\text{location does not improve the model})$$

$$H_A : \log(\lambda) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{age}^2 + \beta_3 \text{location} \quad (\text{location improves the model})$$


```
# Fit quadratic model with location
modela2L <- glm(total ~ age + age2 + location,
               family = poisson, data = fHH1)
summary(modela2L)
```

```
##
## Call:
## glm(formula = total ~ age + age2 + location, family = poisson,
##      data = fHH1)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.3843338   0.1820919  -2.111  0.03480 *
## age            0.0703628   0.0069051  10.190 < 2e-16 ***
## age2          -0.0007026   0.0000642  -10.944 < 2e-16 ***
## locationDavaoRegion -0.0193872   0.0537827  -0.360  0.71849
## locationIlocosRegion  0.0609820   0.0526598   1.158  0.24685
## locationMetroManila  0.0544801   0.0472012   1.154  0.24841
## locationVisayas     0.1121092   0.0417496   2.685  0.00725 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 2362.5 on 1499 degrees of freedom
## Residual deviance: 2187.8 on 1493 degrees of freedom
## AIC: 6574.7
##
## Number of Fisher Scoring iterations: 5
```

```
# Drop-in-deviance test: quadratic vs quadratic + location
drop_in_dev_3 <- anova(modela2, modela2L, test = "Chisq")
drop_in_dev_3
```

```
## Analysis of Deviance Table
##
## Model 1: total ~ age + age2
## Model 2: total ~ age + age2 + location
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      1497      2200.9
## 2      1493      2187.8  4   13.144  0.01059 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Rate ratios and confidence intervals
exp(coef(modela2L))
```

```
##              (Intercept)              age              age2
##              0.6809041              1.0728974              0.9992977
## locationDavaoRegion locationIlocosRegion locationMetroManila
##              0.9807995              1.0628797              1.0559914
## locationVisayas
##              1.1186350
```

```
exp(confint(modela2L))
```

```
## Waiting for profiling to be done...
```

		2.5 %	97.5 %
## (Intercept)		0.4750208	0.9698856
## age		1.0585965	1.0876426
## age2		0.9991708	0.9994223
## locationDavaoRegion		0.8825025	1.0896753
## locationIlocosRegion		0.9585371	1.1783578
## locationMetroManila		0.9629146	1.1586642
## locationVisayas		1.0313164	1.2147232

The fitted model incorporating location is expressed as follows:

$$\log(\text{total}) = -0.384 + 0.070 \cdot \text{age} - 0.00070 \cdot \text{age}^2 + 0.061 \cdot \text{IlocosRegion} + 0.054 \cdot \text{MetroManila} + 0.112 \cdot \text{Visayas} - 0.019 \cdot \text{DavaoRegion}$$

The coefficients for age and age² remained stable in magnitude and statistical significance following the inclusion of location, suggesting that the quadratic relationship between age and household size is robust to the addition of this covariate.

Reading the region coefficients individually:

- **Visayas** ($\hat{\beta} = 0.112$, $p = 0.007$): Households here are about 11.9% larger than Central Luzon, on average, after controlling for age, and this difference is statistically significant.
- **Ilocos Region** ($\hat{\beta} = 0.061$, $p = 0.25$): Slightly larger than Central Luzon, but not statistically significant — this could simply be due to random variation.
- **Metro Manila** ($\hat{\beta} = 0.054$, $p = 0.25$): Similarly slightly larger than Central Luzon, but also not statistically significant.
- **Davao Region** ($\hat{\beta} = -0.019$, $p = 0.72$): Slightly smaller than Central Luzon, but again not statistically significant.

So, when considered individually, only Visayas stands out as truly different from Central Luzon.

When a categorical variable like location creates multiple dummy variables, you can't just look at each one separately and decide if "location matters." You have to test whether location as a whole — all 4 regions together — improves the model. That's exactly what the drop-in-deviance test does.

We compared the model without location to the model with location, and the deviance dropped by 13.14 with a p-value of 0.011 — below 0.05. So even though 3 out of 4 regions weren't individually significant, as a group, location does meaningfully improve the model. Therefore, we reject H_0 and conclude that location, as a whole, is a statistically significant predictor of household size after controlling for age, and it is retained in the final model.